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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/708,653	03/30/2022	Kayoko Matsunaga	129249-01202	8040

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EXAMINER	
DUTT, ADITI	

ART UNIT	PAPER NUMBER
1675	

NOTIFICATION DATE	DELIVERY MODE
07/01/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/708,653	Applicant(s) Matsunaga et al.	
	Examiner ADITI DUTT	Art Unit 1675	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 28 April 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 13-15,17-18 and 21-27 is/are pending in the application.

5a) Of the above claim(s) 21-25 and 27 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 13-15,17-18 and 26 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 30 March 2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 3/30/22;9/7/22.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☑ Other: Appendix a-c.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

Status of Application, Amendments and/or Claims

2. Applicant's response dated 4/28/2026 is considered and entered into record. Claims 13, 15, 17-18 are amended. Claims 1-12, 16, and 19-20 have been cancelled. New claims 21-27 are added. Claims 13-15, 17-18 and 21-27 are currently pending.

Election with traverse

3. Applicant's election with traverse of Group II (claims 13-15, 17-18 and 26); one polypeptide denoted as "(E1) to (E25)"; and the amino acid sequence species of SEQ ID NO: 997 of polypeptide (E23), in the reply filed on 28 April 2026 is acknowledged.

4. The traversal is on the ground(s) that "there is no additional search burden". Applicant argues that since "new claim 21 is directed to a method for detecting binding between an antigen and antibody, which is the same type of invention as claim 13", "Group II also encompasses claim 21 and claims dependent therefrom – claims 22-25 and 27". Applicant therefore, requests the examination of new claims 21-27 with the claims of Group II.

5. Applicant's arguments are fully considered, however, are not found to be persuasive. Applicant was required to elect one polypeptide from (E1) to (E25) polypeptides (i.e. 25 polypeptides) (para 15 of Restriction), in response to which Applicant appears to have elected (E23) polypeptide and Group II invention. New claims 21-25 and 27 are directed to E1 and E2 polypeptides. As stated in para 16 of the restriction requirement, each of the (E1) to (E25) polypeptides are distinct polypeptides representing distinct proteins (Table 1 of the specification), with unique amino acid sequences of different length and encoded by unique nucleic acid sequences. (E23) corresponding to SEQ ID NO: 16 is an Immunoglobulin light chain, lambda gene cluster protein (page 20 of specification), while (E1) and (E2) polypeptides, correspond to Immunoglobulin M heavy chain secretory form (specification - page 71, lines 1-2, 12-13). Therefore, searching all of the polypeptides ((E23), (E1) and (E2)) in a single patent application would provide an undue search burden on the examiner.

6. Applicant's response did not include an election of parts (ii) and (iii) of the restriction requirement (see para 15 of restriction). However, since amended claims 13 and 26 recite SEQ ID NOs: 16 and 15 along with the elected sequence of SEQ ID NO: 997 of (E23 polypeptide), and since SEQ ID NOs: 16 and 15 correspond to the protein and DNA sequences of the (E23) polypeptide (page 20, Table 1-8 of specification), SEQ ID NOs: 16 and 15 are therefore, respectively construed as being elected (as one each from parts (ii) and (iii) of para 15 of the restriction requirement).

7. Based upon the above analysis, the following elections will be considered for current examination:

Group II;

(i) (E23) polypeptide, (ii) SEQ ID NO: 16, (iii) SEQ ID NO: 15 (with reference to para 15 of the action dated 12/29/25);

Amino acid sequence species of SEQ ID NO: 997 (as one of (E23) polypeptide of instant claim 13).

8. Claims 21-25 and 27 are withdrawn from further consideration pursuant to 37 CFR1.142(b), as being drawn to a nonelected invention, there being no allowable generic or linking claim. Election was made with traverse in the reply filed on 28 April 2026.

9. **The requirement is still deemed proper and is therefore made FINAL.**

10. Claims 13-15, 17-18 and 26, drawn to a method for detecting binding between an antigen and antibody in a sample, are considered for examination in the instant application.

Specification

11. The disclosure is objected to because of the following informalities:

A) Internet address:

The disclosure is objected to because it contains an embedded hyperlink and/or other form of browser-executable code (see, for example, page 34, lines 26, 29). Applicant is required to delete the embedded hyperlink and/or other form of browser-executable code. See MPEP § 608.01.

B) The specification has a number "[0090]" at the beginning of a paragraph on page 34, line 22. If this number denotes the paragraph, it is an error as none of other paragraphs of the specification are numbered.

12. Appropriate correction is required.

Claim Rejections - 35 USC § 112-Second paragraph

13. The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

14. The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

15. Claim 17 is rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

16. Claim 17 is unclear for reciting “tester”. The specification teaches a tester composition: comprising an antibody that could be in an immobilized form, OR comprising an antigen, wherein the tester can be used for determining the presence or absence of an antigen OR an IgE antibody in a subject (page 42, lines 5-6, 12-13; page 5, lines 29-38; page 6, lines 25-26). The specification however, does not explicitly define a “tester”. The claim fails to identify the metes and bounds of the related subject matter and how that could be ascertained in the stated invention. For present examination, any composition comprising an antibody for detecting the polypeptide antigen in a sample, will be considered as a “tester”.

17. Appropriate clarification and correction are required.

Claim Rejections - 35 USC § 102

18. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

19. Claims 13-15, 17-18 and 26 are rejected under AIA 35 U.S.C. 102(a)(1) as being anticipated by Okano et al (US PGPB 20110130343, 6/2/2011), and as evidenced by Nieters et al (*Carcinogenesis* 35: 2716-22, 2014), and (Monroy-Iglesias et al, *Clin Exptl Immunol* 209: 46-63, 2022).

20. The claims are directed to a method for detecting binding between an antigen and an antibody, wherein the antigen comprises an epitope that specifically binds to an allergen specific IgE antibody from a patient allergic to the allergen, and the antibody binds to a polypeptide (E23) comprising an amino acid sequence of any one of SEQ ID NOs: 997-1037, comprising contacting (i) the antigen with an antibody sample to determine the presence or absence of IgE antibody; or (ii) the antibody with an allergen sample to determine the presence or absence of the polypeptide; wherein there is at least 70% or 90% identity of: the polypeptide with SEQ ID NO: 16, or a nucleotide sequence encoding the polypeptide with SEQ ID NO: 15, or a part thereof (claims 13, 26); wherein: the antigen or antibody is immobilized to a carrier or a surface (claim 14); and the antigen and antibody are from a kit comprising a reagent for detecting binding between the antigen and antibody (claim 15). Claim 17 recites determining the presence or absence of an antigen in an object of interest (sample), comprising contacting a tester comprising the antibody with a sample comprising the antigen polypeptide (claim

17). Claim 18 recites a method for detecting an allergen polypeptide of claim 13 in a sample using the method recited in claim 13, wherein the antibody is from a kit that optionally further comprises the antigen of claim 13.

21. Okano et al teach a polypeptide (antigen) that can react (bind to) with antibodies in serum of a cancer patient, wherein the polypeptide consists essentially (comprises) an amino acid sequence of SEQ ID NO: 95 that is:100% identical to instant SEQ ID NO: 997 (Appendix a); 90.9% (at least 70%) identical to instant SEQ ID NO: 16 (Appendix b); and SEQ ID NO: 94, which is 89.1% (at least 70%) identical to the nucleotide sequence of instant SEQ ID NO: 15, wherein SEQ ID NO: 94 encodes the amino acid sequence of SEQ ID NO: 95 (Appendix c) (Abstract; para 0149, 0073) (*instant claims 13, 26*). The reference teaches measurement of the antibody (Method 1) or the polypeptide (antigen) (Method 2) in a sample, using well-known immunoassays (ELISA, sandwich immunoassay, based upon the antigen-antibody binding), comprising the polypeptide antigen or antibody respectively, and the needed reagents and label, (para 0120, 0124-0127, 0138, 0139), wherein the polypeptide antigen is immobilized on a solid phase surface (para 0125) (*instant claims 13, 14, 17, 26*). Since the reference teaches known immunoassays for measuring the antigen or antibody in a sample, the "contacting" would be an inherent crucial step for binding, as is also evident from the teachings of the instant specification (page 37, lines 21-34).

22. Although Okano et al do not mention the term "tester", the immunoassay of Method 2 uses the antibody for detecting the polypeptide antigen in a sample, wherein the antibody would necessarily be present in a solution (composition). Furthermore, Okano et al teach preparing an antibody, that would immunologically react with the

antigen polypeptide comprising SEQ ID NO: 95 in a sample (contacting with an antigen containing sample) (para 0127). Since the antibody is prepared and used for testing the presence of the antigen polypeptide in well-known immunoassays, the antibody would inherently be considered to be a tester (*instant claim 17*). It is to be noted that the tester (of instant claim 17) can be any composition comprising an antibody or antigen (see para 16 above). Even though Okano et al do not explicitly mention that the antigen and antibody are from a kit, the reference teaches diagnosis of cancer using immunoassays (like ELISA) for detection of the antigen polypeptide, and antibodies binding to said antigen. The reference teaching therefore, inherently includes the antigen and antibody along with the known immunoassay reagents in a pre-assembled compilation (kit) for necessary detection/diagnosis (*instant claims 15, 18*). Note that the instant specification also refers to diagnostic or detection kit (page 3, lines 1,2; page 6, line 17). A kit (detection/diagnostic) is understood in the relevant art as comprising reagents and materials necessary for the assay.

23. Even though Okano et al do not teach that the antigen polypeptide binds to allergen-specific IgE antibody from a patient, the reference teaches that said antigen binds to antibodies in a cancer patient serum. A significant association of IgE levels and the risk of cancers like chronic lymphocytic leukemia (CLL) is evidenced by Nieters et al (Carcinogenesis 35: 2716-22, 2014; Abstract; Discussion, para 1). Relevant art also evinces the association of serum allergen specific IgE and cancer risk in many types of cancers including leukemia, head and neck cancer, lung cancer, etc. (Monroy-Iglesias et al*, Clin Exptl Immunol* 209: 46-63, 2022; page 50, col 2, para 1, 2). Since Okano et al teach that SEQ ID NO: 95 is expressed in leukemia cells (para 0109), wherein SEQ ID

NO: 95 is identical to instant SEQ ID NO: 997 which binds to IgE, and since there is a significant association of serum IgE levels and leukemia (Nieters et al), SEQ ID NO: 95 would inherently bind to IgE, absent any evidence to the contrary. Okano et al therefore, anticipates the invention.

Conclusion

24. No claims are allowed.

25. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Aditi Dutt whose telephone number is (571)272-9037. The examiner can normally be reached on M-F 9:00am-5:00pm.

26. Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

27. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jeffrey Stucker, can be reached on 571-272-0911. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

28. Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/A. D./
Examiner, Art Unit 1675
24 June 2026

/KIMBERLY BALLARD/
Primary Examiner, Art Unit 1675